BIOLOGICTRAJ · PIPELINE AND SYNTHETIC BENCHMARK · ALL DATA SIMULATED

# Testing the pipeline before the patient data arrive

**PRISM-like PBMCs with planted answers: 8 patients, 3 visits, ~89,000 cells.**

Supervisor: Dr Nazanin Zounemat-Kermani · National Heart and Lung Institute, Imperial College London

## The pipeline

**1 · CHECK** — **Stress screen**
Heat-shock vs immediate-early genes

**2 · PSEUDOBULK** — **Summed counts**
Per sample and cell type

**3 · COUNT MODEL** — **edgeR / limma-voom**
Patient blocking; tests each visit step

**4 · TOPOLOGY** — **LongOmicsTraj**
Shape of change, kept only if significant

**5 · SCORE** — **Against the truth**
Planted shapes known

**SCENARIO A · DRUG**
NF-κB/AP-1 genes fall at 6 months as a **real drug effect**.

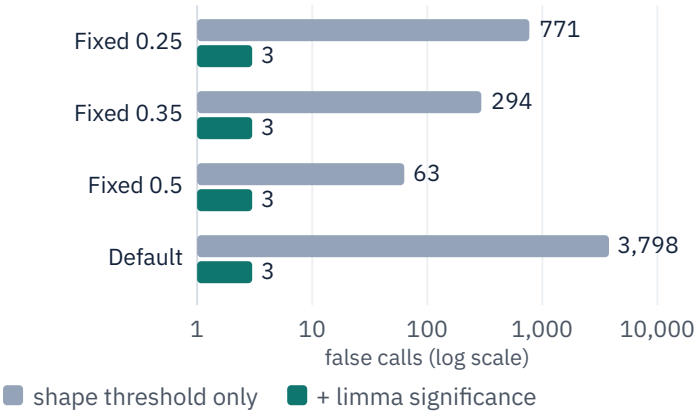
**SCENARIO B · STRESS BATCH**
Same genes fall, but only because 6-month samples were **handled more gently**.

**553 / 553**
planted trajectory shapes recovered (scenario A)

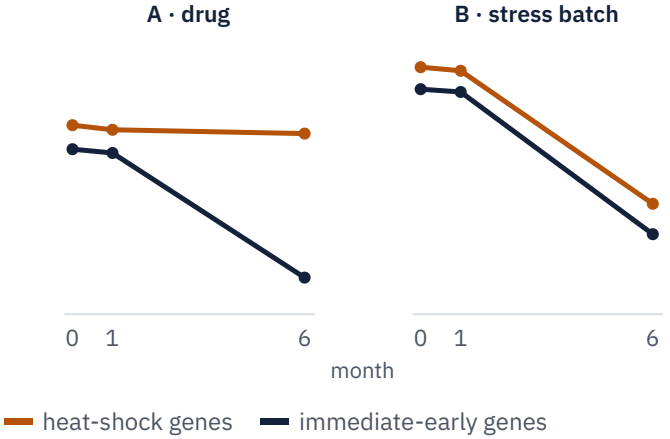
**3** vs 294
false calls in 9,582 unchanged genes, with vs without significance filter

**14 / 14**
cell types flagged by the stress screen in scenario B; none in A

## Significance matters more than the threshold




## Only the stress screen spots the artefact




## What we learned

**1 · DEMAND SIGNIFICANCE**
Keeping a step only when limma supports it removes nearly all false shapes, at any threshold.

**2 · STATISTICS CAN BE FOOLED**
465 of 468 artefactual stress-gene trajectories pass the filter. Heat-shock genes give it away.

**3 · DEPLETED CELLS VANISH**
Anti-IL-5 depletes basophils too far to measure their genes; composition analysis must cover them.

**An upper bound:** no ambient RNA, doublets or batch noise; true cell labels. **Next:** annotation, composition, scGPT and PerturbGen, then PRISM.

Design after Park et al., *Exp Mol Med* 2024 (PRISM). Tools: edgeR, limma, LongOmicsTraj (bioRxiv 2026). Stress genes after van den Brink et al., *Nat Methods* 2017.

CODE
github.com/nzkermani/BiologicTraj

BIOLOGICTRAJ · AIM 1 FIRST PASS AND REVIEW · ALL DATA SIMULATED

# Does a foundation model see more than PCA?

Zero-shot scGPT vs PCA on the same 4,992 synthetic PBMCs, 24 samples.

Supervisor: Dr Nazanin Zounemat-Kermani · National Heart and Lung Institute, Imperial College London

**1 · EMBED**
**Three embeddings**
PCA (50 PCs); scGPT blood and whole-human checkpoints, no retraining

**2 · SCORE**
**Leave one patient out**
Can a new patient's cells be told apart by cell type, and month 6 vs earlier?

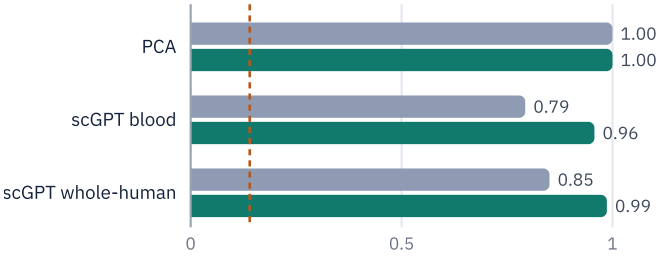
**3 · TWO GENE SETS**
**All 1,625 vs 125 planted**
1,500 filler genes have real names but random expression

**0.85 → 0.99**
scGPT whole-human, cell type: all genes vs planted only. PCA: 1.00 on both.

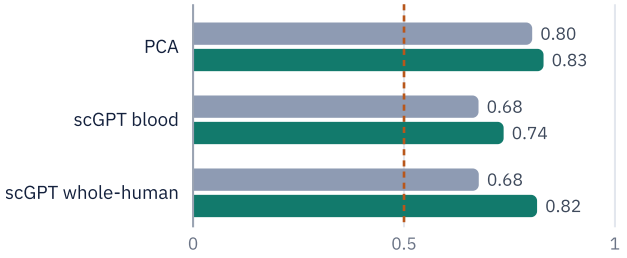
**0.31 → 0.16**
PCA same-patient neighbours (chance 0.13). Patient signal sat in the filler genes.

**< 0.5**
drug-class score for every method, random noise included: the metric itself is biased.

### Cell type accuracy




### Month 6 vs earlier (balanced accuracy)




all 1,625 genes · 125 planted genes only · ----- random-embedding null

## What we learned

**1 · FILLER MISLEADS SCGPT**
Real names, random expression: this misleads a model that reads gene names. Without it, whole-human scGPT matches PCA.

**2 · PATIENT SIGNAL WAS DIFFUSE**
PCA grouped patients by summing small offsets over 1,500 genes. Biology or nuisance? Real data will tell.

**3 · SOME METRICS LIE**
With 4 patients per drug, leaving one out tilts k-NN against its class. Drug needs a patient-level test.

## Review: what to do next

Structured review by Claude in the style of a single-cell methods reviewer and an asthma airway biologist. Tracked as GitHub issues. All of this is synthetic: planted genes are scGPT's best case, and ranking methods needs real data.

1. METHODS **Semi-synthetic benchmark:** inject known effects into real PBMCs. #1
2. BIOLOGY **Write down expected effects** per drug and cell type before modelling. #9
3. BIOLOGY **Rare type-2 cells** (Th2, ILC2, basophils) on all cells, not a subsample. #8
4. METHODS **Patient-level metrics,** scVI/scPoli and reference annotation. #2 #3 #4
5. BIOLOGY **Link to clinical change** and test blood signatures in U-BIOPRED airway data. #10 #11

scGPT: Cui et al., *Nat Methods* 2024 (blood and whole-human checkpoints).
Metrics after scIB, Luecken et al., *Nat Methods* 2022. CPU only, 2–30 min per run.

CODE & ISSUES
github.com/nzkermani/BiologicTraj